

Generator Matrix from Polynomial Shifts

$g(X) = 1 + X + X^3$ generates a cyclic $[7, 4]$ code by shifts.

$$g(X) = 1 + X + X^3$$

1 1 0 1

coefficient vector of g

shift

G

1	1	0	1	0	0	0
0	1	1	0	1	0	0
0	0	1	1	0	1	0
0	0	0	1	1	0	1

g, Xg, X^2g, X^3g

$k = n - \deg g = 4$ rows

Encoding is multiplication $m(X) \mapsto m(X)g(X)$.